

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : SEC-3P

(Mobile App Development)

Full Marks : 50

The figures in the margin indicate full marks.

SET - II

Marks Distribution :

User Interface (UI) / Front End	:	05
Procedure/Steps	:	10
Source Code	:	10
Output	:	07
Conclusion/Discussion	:	03
Experiment Total	:	35
Laboratory Notebook	:	05
Viva voce	:	10
Total	:	50

Answer *any one* question.

1. Develop a Flutter app using Dart programming that calculates the Restaurant Tip amount based on the total bill and the selected Tip percentage. The app should include input fields for users to enter the bill amount, Enter Tip percentage and buttons to calculate the Tip. Ensure that the app displays the calculated Tip amount and the total bill, using Indian currency (INR). 35
2. Develop a Flutter app using Dart programming to calculate the area and circumference of a circle based on the input radius. The app should provide a dropdown list for users to select either the area or circumference calculation. Ensure that the app displays the calculated result. 35

3. Develop a single-page app using Flutter with the following features :
- An AppBar displaying the app name as 'XYZ Tour & Travels.'
 - A side drawer with the name 'Tourist-spots'.
 - A floating action button that, when clicked, navigates to a page displaying pictures of different tourist places. 35
4. Develop a Flutter app using Dart programming that allow to submit 5 random name through text data field and displays same upon clicking a button. Include another button that navigates to a different or same page. 35
5. Develop a Flutter expense tracker app using Dart programming that calculate expense by user-given dates. Display a summary of expenses by category and date. 35
6. Develop a Flutter app using a stateful widget that changes the AppBar name when the user enters a name. The app should include an input field for the user to enter their name, and the AppBar title should update accordingly. 35
7. Develop a Flutter app with a login page that includes the following validation rules :
- (a) The username must be at least 6 characters long.
 - (b) The password must contain :
 - (i) An uppercase letter
 - (ii) A lowercase letter
 - (iii) Digits
 - (iv) A special character
 - (v) More than 6 characters.
- The login button should be displayed and when clicked, it should show any validation errors if present. 35
8. Develop a Flutter app using Dart programming to calculate the volume of a cube and a cylinder. The app should provide text boxes for side, radius and a dropdown to select height of a cylinder. Display the volumes in the same page. Use buttons when pressed will calculate the volume and a clear button which will clear the content. 35